

Title : PCB DEFECT DETECTION USING CONDITIONAL DCGAN

TEAM 10

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Introduction

- Printed Circuit Boards are a part of electronic devices.
- When Printed Circuit Boards have defects it can make the products not work well and it affects how the circuit works.
- Checking Printed Circuit Boards by hand is really hard. It takes a lot of time.
- Using Artificial Intelligence and deep learning techniques can help us find defects in Printed Circuit Boards automatically.
- We can use something called GAN models to make pictures of defective Printed Circuit Boards to help with training and making more data.
- This project is about making Printed Circuit Boards using GAN and then using deep learning techniques to find the defects, in Printed Circuit Boards.

Problem Statement

Detecting PCB defects needs a lot of labeled training data for deep learning results.. Collecting images of PCB defects for different types is hard and limited. This project aims to create synthetic PCB defect images using GAN and improve PCB defect detection, with YOLO-based deep learning techniques.

PCB defect detection is important. Gan helps in generating PCB defect images and YOLO is used for PCB defect detection.

Objectives

- To generate synthetic PCB defect images using GAN
- To detect 6 different types of PCB defects using YOLO
- To analyze PCB defect detection performance using deep learning techniques

Existing Methods Points

- Manual PCB inspection
- Traditional image processing methods
- CNN-based defect detection
- Limited PCB defect datasets
- Time-consuming inspection process

Proposed Method

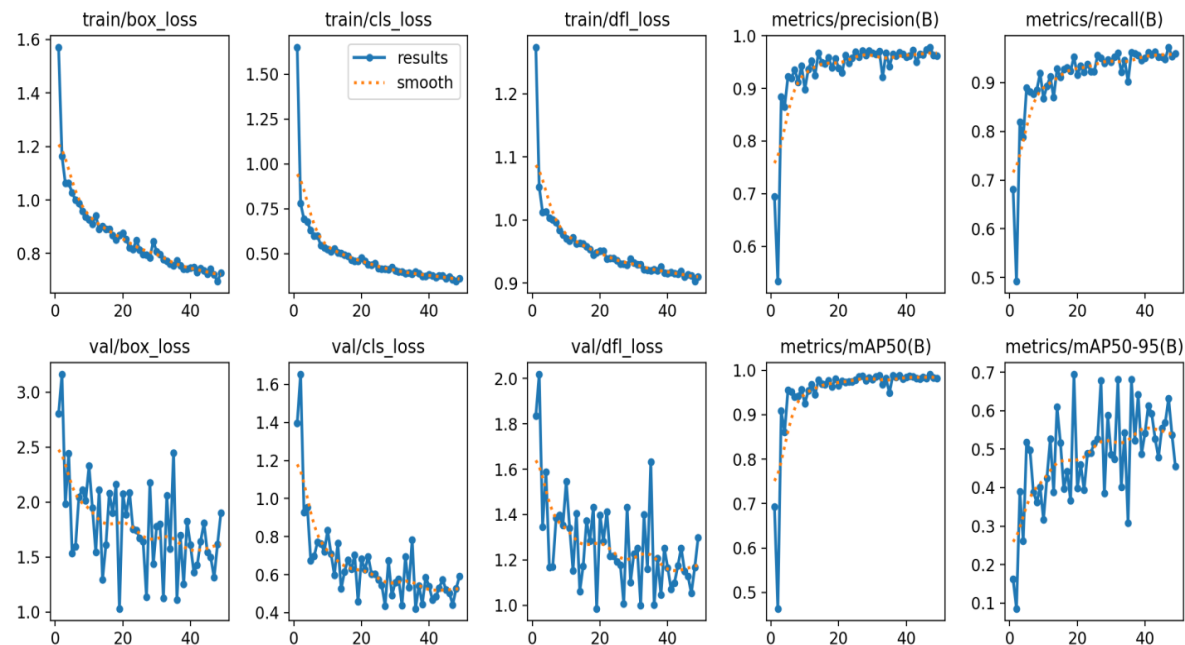
- Deep PCB dataset is used
- GAN is used for synthetic PCB defect image generation
- YOLO is used for PCB defect detection
- Deep learning techniques are used for performance analysis

Implementation

- DeepPCB dataset collected
- Image preprocessing performed
- Labels extracted from TXT files
- Conditional DCGAN used for synthetic PCB defect generation
- Data augmentation applied
- YOLOv12-S model trained using real and synthetic images
- PCB defects detected using trained model
- Performance evaluated using Precision, Recall, and mAP metrics

Results

- High-quality synthetic PCB defect images generated
- Improved dataset balance
- Realistic PCB defect patterns created
- Better detection performance achieved
- Improved PCB defect detection accuracy
- YOLOv12-S successfully detected different PCB defects



Evaluation Metrics

Precision : 98.63%

Recall : 95.28%

mAP@50 : 98.00%

mAP@50-95 : 69.48%



Applications

- PCB manufacturing industry
- AI-based inspection systems
- Automated PCB defect detection
- Quality control automation
- Smart electronics manufacturing
- Training dataset generation for AI models
- Industrial defect monitoring systems
- Real-time PCB inspection systems

Advantages

- Reduces data collection cost
- Solves data scarcity problem
- Improves AI model performance
- Generates realistic synthetic PCB defect images
- Improves dataset diversity and balance
- Supports accurate PCB defect detection
- Generates unlimited training samples
- Enhances automated inspection systems

Limitatons

- GAN training instability
- Requires parameter tuning
- Synthetic images may slightly differ from real PCB textures
- Training process requires high computational power
- Large datasets are needed for better generation quality
- Model performance depends on training quality

THANK YOU